# Proximal Policy Optimisation for Multi-Asset ETF Allocation under Transaction Costs

hypoq autonomous research harness[1]

[1]Hypothesis × Quant

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**Abstract**

**Hypothesis:** $H_0$: the Sharpe ratio of a PPO allocation policy does not exceed that of daily-rebalanced equal weight; $H_1$: it does

**Data:** Total returns of SPY, TLT, GLD, EEM, HYG; training before 2023, out-of-sample 2023-02-01 to 2026-10-22 (935 days)

**Method:** PPO (3 seeds, 120,000 steps each, averaged), softmax weights, 5 bps cost per unit turnover, decision at close $t$ applied to day $t$+1, one-sided stationary-bootstrap test of the Sharpe difference

**Results:** PPO Sharpe 0.18 vs equal weight 1.17 (difference -0.99, $p = 1.000$). Decision: **REJECTED**

## 1 Design

The environment exposes the last 20 daily returns of each asset (scaled by training volatility) and the current drifted weights. The agent outputs unconstrained scores mapped to long-only fully invested weights by a softmax; reward is the log net return. Training episodes are random one-year windows before 2023; the test period is simulated once and never used for tuning [Moody and Saffell, 2001].

## 2 Results

Table 1: Out-of-sample performance net of 5 bps cost per unit turnover

| Strategy | Ann. return | Ann. vol. | Sharpe | Max drawdown |
|---|---|---|---|---|
| PPO ensemble | 1.34% | 11.16% | 0.18 | -15.1% |
| Equal weight | 12.05% | 10.19% | 1.17 | -8.7% |
| SPY buy-and-hold | 20.06% | 14.82% | 1.31 | -18.8% |
| 60/40 SPY/TLT | 10.60% | 11.08% | 0.96 | -12.7% |

The ensemble policy turns over 64.4% of the portfolio per day on average.

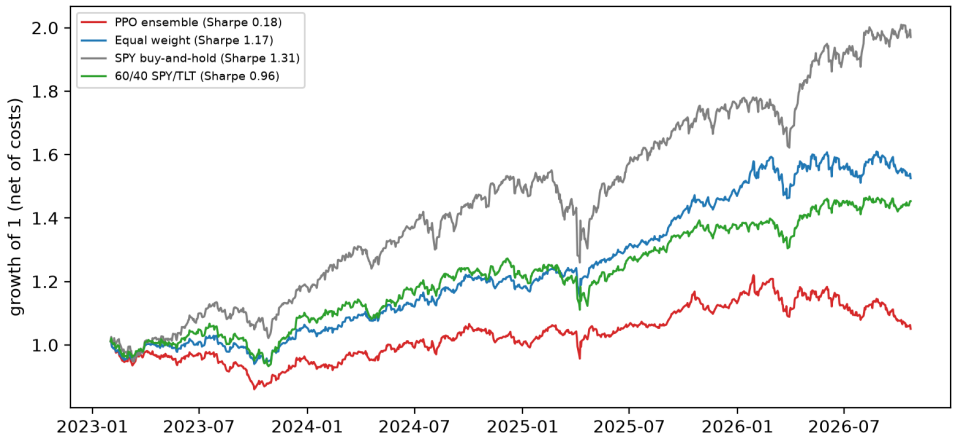



Figure 1: Out-of-sample growth of one unit.

## 3 Limitations

Five ETFs and a single three-year test window give low statistical power; cash and financing are ignored; PPO hyper-parameters were fixed a priori, not searched, and results vary across seeds (averaged here).

## References

John Moody and Matthew Saffell. Learning to trade via direct reinforcement. *IEEE Transactions on Neural Networks*, 12(4):875–889, 2001.